Relative Covariance

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July 28, 2026

Note: *Parts of this document may be AI-written. My general workflow is to write sections of the article myself, laying out the general points I want to hit on and going into detail where relevant. I then converse with an LLM to identify errors in the logic or math and opportunities to use clearer or more generalizable notation. I then have the LLM make the edits to reflect our proposed changes, including adding more detailed derivations.*

This article describes how the phenotypic covariance between two individuals i and j can be modeled by limited information we have about the pair. I derive formulas from the starting trait generative model and only introduce assumptions at the point the derivation requires it. I start with a simple trait architecture and iteratively add more complexity.

1 Additive genetics under random mating with independent environment

We begin with the simplest architecture: a trait determined by an additive genetic component and an independent environmental component, in a randomly mating population.

1.1 The phenotype model

For an individual i we write the (standardized) phenotype as

$$y_i = g_i + e_i, \tag{1}$$

where g_i is the genetic value and e_i the environmental deviation. We take y to be standardized across the population, $\mathrm{E}[y] = 0$ and $\mathrm{Var}[y] = 1$, and both components to be mean-centered, $\mathrm{E}[g] = \mathrm{E}[e] = 0$, with variances $\mathrm{Var}[g] = V_A$ and $\mathrm{Var}[e] = V_E$.

We then introduce the following assumptions.

[Additivity] Additive genetic value

The genetic value is a fixed additive function of the individual's own genotype,

$$g_i = \sum_{k=1}^{M} \beta_k \, x_{ik},$$

where β_k is the per-standardized-allele effect of variant k, x_{ik} is the standardized genotype ($\mathrm{E}[x_{ik}] = 0$, $\mathrm{Var}[x_{ik}] = 1$), and M is the number of variants in the genome. Non-causal variants are included with $\beta_k = 0$.

Commonly bundled into: the additive model; no dominance, no gene–gene interactions (GxG, epistasis), and no gene–environment interactions (GxE) — each would make the genetic value depend on more than the individual's own genotype in an additive way.

[GE-indep] Gene–environment independence

An individual's environmental component is independent of their own genetic component:

$$g_i \perp e_i$$

Commonly bundled into: no gene–environment correlation or interaction, within an individual.

[No-IGE] No indirect genetic effects

One individual's genetic component is independent of another individual's environmental component:

$$g_i \perp e_j \quad (i \neq j).$$

Commonly bundled into: no indirect (e.g. parental) genetic effects — one individual's genotype does not shape another's environment.

[Env-indep] Independent environments

The environmental component is independent across individuals, including relatives:

$$e_i \perp e_j \quad (i \neq j).$$

Commonly bundled into: no shared/common environment (C); no environmental confounder shared by the pair (e.g. population stratification).

Because $g_i \perp e_i$ (**[GE-indep]**), $\mathrm{Cov}[g_i, e_i] = 0$ and the phenotypic variance decomposes as $\mathrm{Var}[y] = V_A + V_E$. Narrow–sense heritability is typically defined as:

$$h^2 = \frac{V_A}{\mathrm{Var}[y]}$$

Since $\mathrm{Var}[y] = 1$, $h^2 = V_A$ under this model.

1.2 Phenotypic covariance is genotypic covariance

For a single fixed pair, y_i and y_j are two fixed values, so "their covariance" only has meaning as an average over an ensemble of pairs. The ensemble we use is the set of all pairs that share a given piece of *relatedness information* $\mathcal{I}_{ij}$ about the pair, and the object of interest is the conditional cross-product

$$C_{ij}(\mathcal{I}) \;:=\; \mathrm{E}[\, y_i y_j \mid \mathcal{I}_{ij}\,]. \tag{2}$$

If $\mathrm{E}[y_i \mid \mathcal{I}] = 0$, then $C_{ij}(\mathcal{I}) = \mathrm{Cov}[y_i, y_j \mid \mathcal{I}]$. For most information $\mathcal{I}$, like the realized relatedness or degree between two relatives, this equivalence is applicable. An exception is when the information we are modelling contains individuals' actual genotype information, for which $\mathrm{E}[y_i \mid \text{genotypes}] = g_i \neq 0$, therefore the expected cross-product of two individuals is not equivalent to their covariance given the fixed information.

Using (1) together with **[No-IGE]** and **[Env-indep]**, every cross term involving e vanishes in expectation (each factors through a mean-zero environmental term), so the phenotypic cross-product reduces to the genetic one:

$$C_{ij}(\mathcal{I}) \;=\; \mathrm{E}[\,(g_i + e_i)(g_j + e_j) \mid \mathcal{I}\,] \;=\; \mathrm{E}[\, g_i g_j \mid \mathcal{I}\,]. \tag{3}$$

The rest of the article evaluates (3) at four levels of resolution for $\mathcal{I}$.

1.3 Level 0 — the exact genetic cross-product

At the finest resolution we condition on the realized genotypes of i and j at the causal variants, for which we know the true causal effect sizes. Then g_i and g_j are fixed, and by **[Additivity]**,

$$C_{ij} \;=\; \mathrm{E}[\, y_i y_j \mid \mathbf{x}_i, \mathbf{x}_j, \boldsymbol{\beta}\,] \;=\; g_i g_j \;=\; \sum_k \sum_{k'} \beta_k \beta_{k'}\, x_{ik}\, x_{jk'}. \tag{4}$$

Under our model and assumptions so far, this returns the true expected cross-product of two individuals' phenotypes. The genetic cross-product between two individuals is, fundamentally, an effect-size-weighted function of the genotypes they actually carry at causal sites. Notably, this formula does not require any knowledge of the relatedness of the two individuals.

1.4 Level 1: genome-wide allelic sharing

If we do not know the causal effect sizes, then we cannot apply (4). However, we can assume that genetic effects' cross-products are independent of individuals' allelic sharing patterns.

[Random-effects] Random effects, unpatterned by sharing

Treating the effects as random, they are mean-zero and uncorrelated across variants, the distribution of their cross-products $\{\beta_k \beta_{k'}\}$ is independent of the genotypes $\mathbf{X}$, and every variant carries the same second moment:

$$\mathrm{E}[\beta_k] = 0, \quad \mathrm{E}[\beta_k \beta_{k'}] = 0 \ \ (k \neq k'), \quad \mathrm{E}[\beta_k \beta_{k'} \mid \mathbf{X}] = \mathrm{E}[\beta_k \beta_{k'}], \quad \mathrm{E}[\beta_k^2] = \sigma_\beta^2.$$

Commonly bundled into: the infinitesimal / random-effects model; causal variants not concentrated in more- or less-shared regions of the genome.

Taking the expectation of (4) over the effects, and using the independence of the cross-products from the genotypes (**[Random-effects]**) to replace $\mathrm{E}[\beta_k \beta_{k'} \mid \mathbf{X}]$ by $\mathrm{E}[\beta_k \beta_{k'}]$:

$$\mathrm{E}[\, g_i g_j \mid \mathbf{x}_i, \mathbf{x}_j \,] = \sum_k \sum_{k'} \mathrm{E}[\beta_k \beta_{k'}]\, x_{ik} x_{jk'}.$$

Because the effects are mean-zero and uncorrelated across variants, $\mathrm{E}[\beta_k \beta_{k'}] = 0$ whenever $k \neq k'$ and σ_β^2 when $k = k'$; the entire off-diagonal ($k \neq k'$) collapses regardless of any correlation between the genotypes. What remains is

$$\mathrm{E}[\, g_i g_j \mid \mathbf{x}_i, \mathbf{x}_j \,] = \sigma_\beta^2 \sum_k x_{ik} x_{jk}.$$

The common second moment is fixed by variance bookkeeping: since $\mathrm{Var}[x_{ik}] = 1$ and the off-diagonal effect terms vanish, $V_A = \mathrm{Var}[g_i] = \sum_k \mathrm{E}[\beta_k^2] = M\sigma_\beta^2$, hence $\sigma_\beta^2 = V_A/M$. Therefore

$$C_{ij}(\mathbf{x}_i, \mathbf{x}_j) \ = \ V_A \cdot \frac{1}{M} \sum_k x_{ik} x_{jk}. \tag{5}$$

The per-variant effect weighting of (4) has been replaced by the genome–wide genotype cross-product (i.e. allelic sharing), which we'll define by $S_{ij} = \frac{1}{M} \sum_k x_{ik} x_{jk}$. This can be thought of as an identity-by-state (IBS)-based measure of "relatedness". Therefore, we can further simplify:

$$C_{ij}(S_{ij}) \ = \ V_A S_{ij}. \tag{6}$$

1.5 Level 2 — realized relatedness

The problem with calculating S_{ij} is that we may not have genotyped every causal variant that contributes to a trait. As long as some of the causal variants are not in perfect correlation with a genotyped variant, then the covariance of two individuals' phenotypes will not fully take into account the contributions of the ungenotyped causal variants. However, when individuals are related, their IBS-based relatedness from genotyped variants is informative of the IBS-based relatedness of ungenotyped variants. Therefore, the former can be used to capture the contributions of the latter.

When individuals inherit a genomic segment from a common ancestor, then all of the variants within that genomic segment will be identical-by-state, as long as we assume that de-novo mutations have not caused the inherited alleles to be different. These genomic segments and their variants are said to be identical-by-descent (IBD).

[No-mutation] IBD implies IBS

Alleles that are identical-by-descent are also identical-by-state: descent is transmitted without mutation.

Commonly bundled into: the infinite-sites / no-recurrent-mutation assumption.

When two individuals' alleles are IBD, then they are always IBS; when they are not IBD, they can still be IBS. Therefore, we can decompose the IBS-based relatedness S_{ij} into a sum over IBD allele pairs and a sum over non-IBD allele pairs. Write each standardized genotype as the sum of its two per-allele (maternal and paternal) standardized values, $x_{ik} = z_{ik}^{(m)} + z_{ik}^{(p)}$. Since $\mathrm{Var}[x_{ik}] = 1$ and the two alleles are independent ([No-inbreeding]), each per-allele value has $\mathrm{E}[z] = 0$ and $\mathrm{Var}[z] = \frac{1}{2}$. The genotype cross-product then expands over the four ordered allele pairs, and S_{ij} splits by IBD status:

$$\begin{aligned} S_{ij} &= \frac{1}{M}\sum_k x_{ik}x_{jk} = \frac{1}{M}\sum_k \sum_{u,v\in\{m,p\}} z_{ik}^{(u)} z_{jk}^{(v)} \\ &= \underbrace{\frac{1}{M}\sum_{\text{IBD pairs}} z_{ik}^{(u)} z_{jk}^{(v)}}_{\text{IBD}} + \underbrace{\frac{1}{M}\sum_{\text{non-IBD pairs}} z_{ik}^{(u)} z_{jk}^{(v)}}_{\text{non-IBD}} . \end{aligned} \tag{7}$$

For an IBD allele pair, the two per-allele values are copies of the same ancestral allele and, by [No-mutation], of the same state; they are therefore equal, so their expected cross-product is $\mathrm{E}[(z_{ik}^{(u)})^2] = \mathrm{Var}[z_{ik}^{(u)}] = \frac{1}{2}$. For non-IBD allele pairs, we have to make an assumption on how they are distributed.

[Non-IBD-indep] Independence of non-IBD alleles

Any two alleles that are *not* IBD are independent draws from the population allele distribution.

Commonly bundled into: between individuals: random mating, no population structure, and no assortative mating.

[No-inbreeding] No inbreeding

Within an individual, the two homologous alleles are not IBD with each other, and hence (by [Non-IBD-indep]) independent. This results in each variant's genotype variance being $2p_k(1-p_k)$ — so the standardized x_{ik} has variance 1 — and it keeps the number of alleles a pair shares IBD at a variant in $\{0,1,2\}$.

Commonly bundled into: a randomly mating, outbred population (Hardy–Weinberg within individuals).

By the definition of independent draws, the expected cross-product of a non-IBD allele pair is therefore $\mathrm{E}[z_{ik}^{(u)}]\,\mathrm{E}[z_{jk}^{(v)}] = 0$. Letting $\mathrm{IBD}_{ij}(k) \in \{0,1,2\}$ denote the number of alleles the pair shares IBD at variant k, exactly $\mathrm{IBD}_{ij}(k)$ of the four ordered allele pairs are IBD — each contributing $\frac{1}{2}$ — and the rest contribute 0, so

$$\mathrm{E}[\,x_{ik}x_{jk} \,|\, \mathrm{IBD}_{ij}(k) = s\,] = s \cdot \tfrac{1}{2} = \frac{s}{2} \qquad (s = 0, 1, 2).$$

A variant thus contributes expected cross-product 0, $\frac{1}{2}$, or 1 according to whether the pair shares 0, 1, or 2 alleles IBD there.

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We now evaluate the per-variant cross-product $\mathrm{E}[x_{ik}x_{jk} \mid \mathcal{S}]$ in terms of IBD. It is cleanest to work at the level of individual alleles. At variant k (with population allele frequency p_k), let individual i carry the two allele indicators $a_{ik}^{(m)}, a_{ik}^{(p)} \in \{0,1\}$ (maternal and paternal), each Bernoulli(p_k), so that the allele dosage is $G_{ik} = a_{ik}^{(m)} + a_{ik}^{(p)}$ and

$$x_{ik} = \frac{G_{ik} - 2p_k}{\sqrt{2p_k(1-p_k)}}.$$

Two alleles are IBD if they descend from — and are copies of — the same ancestral allele. Let $\mathrm{IBD}_{ij}(k) \in \{0,1,2\}$ denote the number of alleles individual i shares IBD with j at variant k.

Under [No-inbreeding] the two alleles within an individual are independent, so $\mathrm{Var}[G_{ik}] = 2p_k(1-p_k)$ and the standardization indeed gives $\mathrm{Var}[x_{ik}] = 1$. For the pair, conditioning on descent leaves the marginal allele distributions unchanged, so $\mathrm{E}[x_{ik}] = 0$ and $\mathrm{E}[x_{ik}x_{jk} \mid \mathcal{S}] = \mathrm{Cov}[G_{ik}, G_{jk} \mid \mathcal{S}]/\big(2p_k(1-p_k)\big)$. Expanding the genotype covariance over the four ordered allele pairs,

$$\mathrm{Cov}[G_{ik}, G_{jk} \mid \mathcal{S}] = \sum_{u\in\{m,p\}} \sum_{v\in\{m,p\}} \mathrm{Cov}[a_{ik}^{(u)},\, a_{jk}^{(v)}],$$

each term is $\mathrm{Var}[a] = p_k(1-p_k)$ if that allele pair is IBD (by [No-mutation] it is the same copy, correlation 1) and 0 if it is not (by [Non-IBD-indep], independent). If the pair shares $\mathrm{IBD}_{ij}(k) = s$ alleles IBD, then exactly s of the four terms are nonzero, giving $\mathrm{Cov}[G_{ik}, G_{jk} \mid s] = s\,p_k(1-p_k)$ and hence the key per-variant identity

$$\mathrm{E}[\,x_{ik}x_{jk} \,|\, \mathrm{IBD}_{ij}(k) = s\,] = \frac{s}{2} \qquad (s = 0, 1, 2). \tag{8}$$

That is, a variant contributes correlation 0, $\frac{1}{2}$, or 1 according to whether the pair shares 0, 1, or 2 alleles IBD there.

To turn the causal-variant average in (6) into a genome-wide quantity we need one more assumption.

[Homog-sharing] Homogeneous genome sharing

The expected IBD sharing is the same at the causal variants as it is genome-wide; IBD is not enriched at particular loci.

Commonly bundled into: causal variants are representative of the genome with respect to relatedness. (Under the all-variants, homoscedastic form of **[Random-effects]** this is already implied — kept here only until Level 2 is revised.)

Substituting (8) into (6) and applying **[Homog-sharing]**,

$$\frac{1}{M}\sum_k \mathrm{E}[\, x_{ik}x_{jk} \mid \mathcal{S}\,] = \frac{1}{M}\sum_k \frac{\mathrm{IBD}_{ij}(k)}{2} \xrightarrow{\textbf{[Homog-sharing]}} \pi_{ij},$$

where π_{ij} is the *realized relatedness* — the proportion of the pair's genome that is IBD. We arrive at the central Level-2 result:

$$\boxed{\mathrm{E}[\, y_i y_j \mid \pi_{ij}\,] \;=\; \pi_{ij}\, V_A}. \tag{9}$$

The phenotypic covariance between two relatives is their realized fraction of shared genome times the additive variance. Because different pairs of the same nominal relationship share different realized fractions (full siblings scatter around $\frac{1}{2}$, and so on), (9) is expressed through the pair's *actual* π_{ij}.

1.6 Level 3 — pedigree degree

Often we do not know π_{ij} but only the pair's pedigree relationship (parent–offspring, full siblings, first cousins, . . .).

[Mendelian] **Mendelian pedigree**

The relationship degree determines the distribution of IBD sharing through Mendelian segregation, so that $\mathrm{E}[\pi_{ij} \mid \text{degree}]$ is the pedigree (expected) relatedness.

Commonly bundled into: a known pedigree with Mendelian inheritance.

Applying the tower property to (9) and averaging over the Mendelian distribution of π_{ij} given the degree d,

$$\mathrm{E}[\, y_i y_j \mid d\,] \;=\; \mathrm{E}\big[\,\mathrm{E}[y_i y_j \mid \pi_{ij}]\,\big|\, d\,\big] \;=\; V_A\,\mathrm{E}[\pi_{ij} \mid d] \;=\; r_d\, V_A, \tag{10}$$

where $r_d = \mathrm{E}[\pi_{ij} \mid d] = 2\phi_{ij}$ is the expected relatedness (twice the kinship coefficient). This recovers the classical covariances between relatives:

Relationship	**Expected relatedness** r_d
Monozygotic twins	1
Parent–offspring	1/2
Full siblings	1/2
Half sibs / grandparent–grandchild / avuncular	1/4
First cousins	1/8
“Unrelated”	0

The step from (9) to (10) is exactly the realized-versus-expected distinction: Level 2 uses a pair's true realized relatedness, whereas Level 3 replaces it with its pedigree expectation, and is therefore only as good as the pedigree and the (often large) sampling variance of π_{ij} around r_d allow — a gap that widens for distant relationships.

1.7 Summary of the tower

Each level conditions on coarser information than the one above and is its conditional expectation; the rightmost column lists the assumption(s) newly required to descend one step.

Level	Conditioned on	$\mathrm{E}[y_i y_j \mid \cdot]$	New assumptions
0	causal genotypes	$g_i g_j$	
			[Additivity] (given **[No-IGE]**, **[Env-indep]**)
1	realized sharing $\mathcal{S}$	$V_A \frac{1}{M} \sum_k \mathrm{E}[x_{ik} x_{jk} \mid \mathcal{S}]$	
			[Random-effects]
2	realized relatedness π_{ij}	$\pi_{ij} V_A$	
			[No-mutation], **[Non-IBD-indep]**, **[No-inbreeding]**
3	pedigree degree d	$r_d V_A$	
			[Mendelian]

Next architectures restart from Level 0 and re-introduce a subset of the assumptions above, relaxing others: e.g. a shared environment breaks **[Env-indep]**; dominance/epistasis breaks **[Additivity]**; assortative mating and population structure break **[Non-IBD-indep]** (correlating non-IBD alleles) and drive a wedge between realized and pedigree relatedness.

References